



REPUBLIC OF THE PHILIPPINES
DEPARTMENT OF EDUCATION
Region VII, Central Visayas
Division of Lapu-Lapu City
SCIENCE AND TECHNOLOGY EDUCATION CENTER
SENIOR HIGH SCHOOL
Basak, Lapu-Lapu City
S.Y. 2026-2027

FIRST SUMMATIVE TEST IN GENERAL SCIENCE- TERM 1

Directions: Read each question carefully and choose the best answer. Shade completely the circle corresponding to your answer (A, B, C, or D) on the provided Answer Sheet. Only the answers shaded on the Answer Sheet will be checked.

1. Which branch of physics primarily explains how a microwave oven heats food?
A. Mechanics B. Thermodynamics C. Electromagnetism D. Nuclear Physics
2. Which branch of physics primarily explains the operating principle of a loudspeaker?
A. Acoustics B. Electromagnetism C. Thermodynamics D. Mechanics
3. A student says, "A thermos keeps drinks hot because it produces heat inside the container." Which statement best corrects this misconception?
A. A thermos generates heat through chemical reactions.
B. A thermos keeps drinks hot by reducing heat transfer to the surroundings.
C. A thermos reflects heat only through conduction.
D. A thermos absorbs heat from the surrounding air.
4. During a class discussion, a learner says, "Noise-canceling headphones work because they completely block all sound from entering the ears." Which statement best explains why this idea is incorrect?
A. They increase the loudness of outside sounds.
B. They produce sound waves that interfere with incoming noise, reducing its intensity.
C. They convert sound into electrical energy.
D. They absorb all sound waves before they reach the ears.
5. A student claims that an air conditioner cools a room by creating cold air. Which statement best explains why this idea is a misconception?
A. An air conditioner creates cold air through a chemical reaction.
B. An air conditioner removes thermal energy from the indoor air and transfers it outside.
C. An air conditioner converts sound energy into thermal energy.
D. An air conditioner lowers the room temperature by reducing gravity.
6. A learner says that an airbag protects passengers because it makes the impact force disappear during a collision. Which statement best corrects this misconception?
A. An airbag eliminates the force acting on the passenger during a collision.
B. An airbag increases the passenger's momentum to prevent injury.
C. An airbag increases the time over which the passenger comes to a stop, reducing the average impact force.
D. An airbag prevents inertia from acting on the passenger.
7. A student says, "*Quantum mechanics is only useful for scientists working in laboratories and has no applications in everyday life.*" Which statement best corrects this misconception?
A. Quantum mechanics is only used to explain the motion of planets and stars.
B. Quantum mechanics has practical applications in technologies such as lasers, semiconductors, and magnetic resonance imaging (MRI).
C. Quantum mechanics applies only to nuclear power plants.
D. Quantum mechanics explains only the behavior of large objects moving at high speeds.



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8. Marie Curie's pioneering research on radioactivity has led to which modern application that improves health care?
 - A. MRI scans using magnetic fields
 - B. Radiation therapy for treating certain types of cancer
 - C. Ultrasound imaging using sound waves
 - D. Laser eye surgery
9. Which statement correctly distinguishes translational motion from rotational motion?
 - A. Translational motion involves movement about an axis, while rotational motion involves movement from one location to another.
 - B. In translational motion, every point of the object experiences the same displacement; in rotational motion, different points travel different distances.
 - C. Rotational motion requires force, whereas translational motion does not.
 - D. Translational motion cannot occur simultaneously with rotational motion.
10. A rolling wheel moves without slipping. Which statement is correct?
 - A. Every point on the wheel has the same linear velocity.
 - B. The wheel undergoes rotational motion only.
 - C. The center of mass undergoes translational motion while the wheel rotates about its center.
 - D. The wheel experiences no angular velocity.
11. Which pair represents rotational counterparts of mass and force?
 - A. Velocity and displacement
 - B. Torque and moment of inertia
 - C. Moment of inertia and torque
 - D. Angular velocity and angular acceleration
12. Two disks have the same mass but different radii. Which has the greater moment of inertia?
 - A. The smaller disk
 - B. The larger disk
 - C. Both have the same moment of inertia.
 - D. Cannot be determined.
13. During a pirouette, a ballet dancer pulls her arms closer to her body. As a result,
 - A. angular velocity decreases because moment of inertia decreases.
 - B. angular velocity increases because moment of inertia decreases.
 - C. torque increases.
 - D. translational velocity increases.
14. Which ergonomic design best reduces torque acting on the lower back while lifting?
 - A. Holding the load close to the body
 - B. Holding the load with extended arms
 - C. Twisting while lifting
 - D. Lifting rapidly
15. Which quantity has no direct rotational counterpart?
 - A. Linear displacement
 - B. Linear velocity
 - C. Linear acceleration
 - D. Mass density
16. A student walks forward while rotating a jump rope. Which statement is correct?
 - A. Only translational motion occurs.
 - B. Only rotational motion occurs.
 - C. Both translational and rotational motions occur simultaneously.
 - D. Neither motion occurs.
17. Why do large steering wheels require less effort than smaller steering wheels?
 - A. Larger radius produces greater torque for the same applied force.
 - B. Larger wheels reduce inertia.
 - C. Larger wheels increase friction.
 - D. Larger wheels reduce angular displacement.
18. A cyclist doubles the radius of the wheel while maintaining the same angular velocity. What will happen to the bicycle's linear speed?
 - A. decrease by half.
 - B. remain unchanged.
 - C. double.
 - D. quadruple.
19. Which statement best distinguishes a compound machine from a simple machine?
 - A. A compound machine contains two or more interacting simple machines.
 - B. A compound machine always has greater efficiency.
 - C. A simple machine always requires electricity.
 - D. Compound machines eliminate friction.
20. A mechanic uses a longer wrench to loosen a bolt. Which principle explains why less force is required?
 - A. Increased mechanical efficiency
 - B. Increased torque due to a longer lever arm
 - C. Reduced gravitational force
 - D. Reduced moment of inertia
21. A lever has an effort arm three times longer than its resistance arm. What is its ideal mechanical advantage?
 - A. 1/3
 - B. 3
 - C. 6
 - D. 9



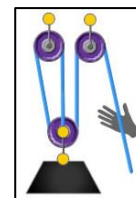
22. Look at the pulley system shown on the right. Based on the number of supporting rope segments holding the load, what is the ideal mechanical advantage (IMA) of the pulley system?

A. 2

B. 3

C. 4

D. 1



23. A screw jack has a handle radius of 0.20 m and a pitch of 0.004 m. Which expression gives its ideal mechanical advantage?

A. $2\pi(0.20)(0.004)$

B. $0.004/2\pi(0.20)$

C. $2\pi(0.20)/0.004$

D. $0.20/0.004$

24. Which modification would increase the ideal mechanical advantage of a screw?

A. Increase its pitch.

B. Decrease its handle radius.

C. Decrease its pitch.

D. Increase friction.

25. A worker uses a 2.5-m long inclined plane to raise a 0.5-m-high platform. The load weighs 480 N, and the required effort is 80 N. What is the efficiency of the inclined plane?

A. 60%

B. 75%

C. 80%

D. 125%

26. Which of the following does NOT belong to the group?

A. Wheelbarrow

B. Nutcracker

C. Bottle opener

D. Tweezers

27. Why do Engineers primarily lubricate moving machine parts?

A. increase torque.

B. reduce friction and energy loss.

C. increase weight.

D. increase mechanical advantage.

28. According to the physics definition of work, which of the following situations does not describe work?

A. A student pushes a stalled car forward along a straight road.

B. A construction worker lifts a box from the floor onto a table.

C. A traveler walks across a hallway while carrying a suitcase at a constant height.

D. A child pulls a sled across a snowy field using a rope.

29. A balloon-powered car travels farther after reducing the mass of its chassis and minimizing axle friction. Which explanation best accounts for the improvement?

A. Greater friction increased acceleration.

B. Lower inertia and reduced energy losses improved motion.

C. The wheels experienced less torque.

D. Gravity increased.

30. A student claims that a machine is more efficient because it finishes a task faster. Which statement best corrects this misconception?

A. A machine is more efficient if it requires less time to complete a task, regardless of the energy used.

B. A machine is more efficient if it converts a greater fraction of the input work into useful output work, even if it does not finish the task faster.

C. A machine is more efficient if it has a greater mechanical advantage because it always requires less effort.

D. A machine is more efficient if it produces a greater output force than the input force.

"Science is powered by desire."

— Senku Ishigami, Dr. STONE



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SECOND SUMMATIVE TEST IN GENERAL SCIENCE-TERM 1

Directions: Read each question carefully and choose the best answer. Shade completely the circle corresponding to your answer (A, B, C, or D) on the provided Answer Sheet. Only the answers shaded completely on the Answer Sheet will be checked.

1. What property of a fluid allows hydraulic systems to transmit force effectively?
A. It can be easily compressed. B. It has no mass.
C. It is nearly incompressible. D. It always flows upward.
2. In a hydraulic system, what happens to the pressure if the cross-sectional area is decreased?
A. increases B. decreases C. remains the same D. cannot be determined

For items 3-4, refer below.

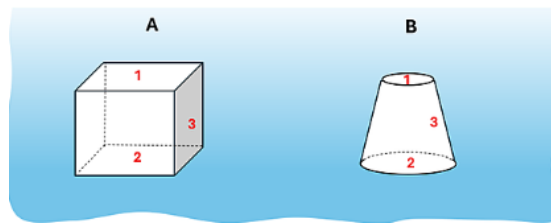
A sphygmomanometer is a special manometer used to measure blood pressure. When a rubber bulb is squeezed, it forces air into the arm cuff until the blood flowing through the artery is stopped. When the valve on the bulb is opened, the cuff pressure starts to drop. When the cuff pressure drops just below the maximum pressure (systolic) produced by the heart, the artery opens again momentarily. The manometer pressure at this point is about 120 mm of mercury for a normal heart. As the pressure in the cuff continues to decrease, a point will be reached where the blood will flow continuously through the artery even when the minimum pressure (diastolic) is produced by the heart. The manometer pressure at this minimum point is about 80 mm of mercury for a normal heart.



3. Why does a nurse usually measure blood pressure around a person's upper arm?
A. The upper arm is closest to the lungs.
B. The upper arm is at the same height as the heart.
C. The upper arm has the largest muscles in the body.
D. The upper arm has the lowest pressure in the body.
4. If blood pressure were measured around the lower leg instead of the upper arm, how would the reading be affected?
A. It would be lower because the leg is farther from the heart.
B. It would be higher because the leg is lower than the heart.
C. It would be equal because pressure is the same throughout the body.
D. It would be unchanged because height does not affect blood pressure.
5. For the shaped container below, which side has the greatest pressure pushing in on it?



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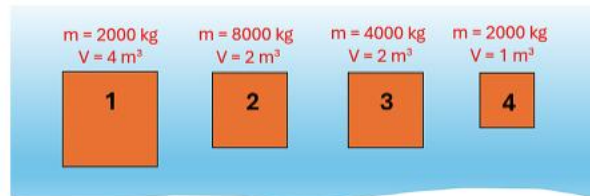


- A. 1 B. 2 C. 3 D. 1,2,3

6. An open tank contains water up to a depth of 350 cm and above it an oil of specific gravity 0.65 for a depth of 2.5 m. What is the pressure intensity at the extreme bottom of the tank?

- A. 5.027 N/cm^2 B. 10.05 N/cm^2 C. 2.51 N/cm^2 D. None of the mentioned

7. Four objects are placed in the same fluid as shown. The masses and volumes of the objects are given. Rank the buoyant force on each from greatest to least.



- A. $1 = 2 = 3 = 4$ B. $1 < 2 = 3 < 4$ C. $1 > 2 = 3 < 4$ D. $1 > 2 = 3 > 4$

8. The density of lead is greater than that of iron, and both metals are denser than water. Two solid objects made of lead and iron have the same dimensions and are completely submerged in water. How does the buoyant force acting on the lead object compare with that acting on the iron object?

- A. The buoyant force on the lead object is greater.
B. The buoyant force on the lead object is equal.
C. The buoyant force on the lead object is less.
D. The buoyant force on the lead object is zero.

9. A questionable merchant sells you a 1.00-kg gold nugget for a very reasonable price. Wondering if you got a deal or were ripped off, you lower the nugget into water and measure the volume of the displaced water (density is 0.00100 kg/cm^3). You find the volume to be 51.8 cm^3 . The density of gold is 0.0193 kg/cm^3 . Did you get the real thing?

- A. Yes B. No C. Maybe D. I don't know

10. From item no. 9, what is the buoyant force acting on this nugget when submerged in the water? Assume $g = 9.8 \text{ m/s}^2$.

- A. 0.098 N B. 0.193 N C. 0.51 N D. 9.80 N

11. Which viscosity is generally most suitable for a hydraulic fluid to ensure efficient machine performance?

- A. A viscosity that allows smooth flow and efficient pressure transmission.
B. A viscosity that makes the fluid flow as slowly as possible.
C. A viscosity that prevents the fluid from moving through the system.
D. A viscosity that increases the mechanical advantage of the machine.

12. A hydraulic press uses a small piston with an area of 5 cm^2 and a large piston with an area of 100 cm^2 . If the input piston moves downward by 20 cm, approximately how far does the output piston move upward in an ideal system?

- A. 1 cm B. 5 cm C. 20 cm D. 400 cm

13. A hydraulic jack allows a person to lift a heavy vehicle using a relatively small input force. Which combination BEST explains how this is achieved?

- A. Equal pressure and different piston areas
B. Equal force and different fluid densities
C. Increased pressure and equal piston areas
D. Reduced pressure and equal piston volumes

14. Which statement BEST explains the relationship between mechanical advantage and piston area in an ideal hydraulic machine?

- A. Mechanical advantage depends on the ratio of output to input piston areas.
B. Mechanical advantage depends only on the density of the hydraulic fluid.



- C. Mechanical advantage depends only on the viscosity of the hydraulic fluid.
 D. Mechanical advantage depends on the total volume of hydraulic fluid.
15. Two pieces of modeling clay have the same mass. Piece A is shaped into a compact ball, while Piece B is shaped into a hollow boat. Which is most likely to float, and why?
- A. Piece A, because compact objects always float better.
 B. Piece B, because its hollow shape allows it to displace more water and have a lower average density.
 C. Both will sink because they have the same mass.
 D. Piece A, because it displaces more water than Piece B.
16. A 100-g piece of clay shaped into a small ball sink in water. If the same clay is reshaped into a larger hollow boat, what will most likely happen?
- A. It will float because its shape increases its volume and allows it to displace more water.
 B. It will sink faster because its mass remains the same.
 C. It will float because its mass has decreased.
 D. It will behave the same because shape and volume do not affect floating.
17. Two identical-sized clay balls are placed in water. Ball A has a mass of 50 g, while Ball B has a mass of 100 g. What will most likely happen?
- A. Ball A is more likely to float because it has a lower density than Ball B.
 B. Ball B is more likely to float because it has greater mass.
 C. Both will float because they have the same volume.
 D. Both will behave exactly the same because shape is the only factor affecting floating.
18. A student uses the same 50-g piece of clay to make two models: Model A has a volume of 20 cm³ and Model B has a volume of 60 cm³. Which model is more likely to float in water?
- A. Model A, because its smaller volume makes it lighter.
 B. Model B, because the same mass spread over a larger volume gives it a lower average density.
 C. Both models will sink because they are made of clay.
 D. Both models will float because they have the same mass.
19. At home, a student plugs a rice cooker, electric kettle, and microwave oven into one extension cord. After several minutes, the extension cord becomes very hot. What is the most likely hazard?
- A. Overloading, which can cause excessive current and overheating of the wires
 B. Open circuit, which stops all current from flowing
 C. High resistance, which prevents the appliances from using electricity
 D. Low voltage, which automatically protects the extension cord
20. Why are household appliances such as lamps and electric fans connected in parallel circuits?
- A. Each appliance can receive the supply voltage and operate independently.
 B. Each appliance must share the same current.
 C. Turning off one appliance automatically turns off all others.
 D. Parallel circuits prevent all possible electrical hazards.
21. A 120 V appliance normally has a resistance of 60 Ω . A fault causes its resistance to decrease to 20 Ω while the voltage remains 120 V. What happens to the current?
- A. It decreases from 2 A to 1 A.
 B. It increases from 2 A to 6 A.
 C. It remains at 2 A.
 D. It becomes zero.
22. Why can a short circuit be dangerous?
- A. It creates a very high resistance that stops all current.
 B. It creates a very low-resistance path that can cause a very large current and overheating.
 C. It always reduces the voltage of the entire power supply to zero.
 D. It prevents electrical energy from being converted into heat.
23. A phone charger becomes unusually hot while charging. What is the best explanation?
- A. The charger may have incorrect electrical specifications, causing overheating.
 B. The charger has correct specifications, so the heat is expected.
 C. The battery has higher resistance, causing less current to flow.
 D. The charger converts electrical energy into heat during normal charging.



24. A circuit breaker trips whenever several high-power appliances are operated at the same time. What is the most likely explanation?
- A. The circuit is drawing excessive current, so the circuit breaker interrupts the circuit as a safety measure.
 - B. The circuit has no resistance, so no current can flow.
 - C. The circuit breaker increases the voltage to make the appliances work faster.
 - D. The appliances have automatically changed from parallel to series connections.
25. A student unplugs a charger by pulling the cord instead of the plug. Why is this unsafe?
- A. It may damage insulation and expose live wires.
 - B. It may increase circuit resistance and reduce current.
 - C. It may increase outlet voltage and damage appliances.
 - D. It may reduce current and affect device performance.
26. A person removes the third prong of a refrigerator's plug. Why is this unsafe?
- A. It removes a safety path for fault current to the ground.
 - B. It adds a path that increases current during normal operation.
 - C. It removes a power connection needed for normal voltage.
 - D. It adds resistance that reduces current during operation.
27. A person touches a live wire and cannot let go. Why can this happen?
- A. Current can cause involuntary muscle contraction.
 - B. Current can increase resistance and constrict the muscles.
 - C. Current can create magnetism that pulls the hand.
 - D. Current can electrically attach the hand to the wire.
28. During a brownout, classroom lights become dim and appliances operate poorly. What does this indicate?
- A. The supply voltage has decreased below its normal level.
 - B. The supply voltage has increased above its normal level.
 - C. The resistance of all appliances has decreased significantly.
 - D. The current has completely stopped flowing through the appliances.
29. A loose outlet produces sparks and crackling sounds when a charger is plugged in. What should be done?
- A. Continue using it if the charger still works normally.
 - B. Stop using it and have it inspected by a qualified electrician.
 - C. Plug in another device to check if the sparks continue.
 - D. Cover the outlet with tape to prevent further visible sparks.
30. A 110 V-only appliance without a built-in voltage converter is plugged directly into a 220 V outlet. What is most likely to happen?
- A. The appliance may be damaged by the excessive voltage.
 - B. The appliance may operate normally at the higher voltage.
 - C. The appliance may automatically reduce the voltage to 110 V.
 - D. The outlet may automatically adjust its voltage to match the appliance.

"I have not failed. I've just found 10,000 ways that won't work."
— Thomas Edison



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